```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load the image
from google.colab import files

uploaded = files.upload()

image_name = next(iter(uploaded))

img = cv2.imread(image_name, cv2.IMREAD_GRAYSCALE)

if img is None:
    print("Error: Image could not be loaded.")
else:
    print("Image loaded successfully.")
    print("Image size:", img.shape)
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 Roll No.: 31


# Convert to grayscale
gray_image = img
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 Roll No.: 31


# Contrast Stretching
min_pixel = np.min(gray_image)
max_pixel = np.max(gray_image)

contrast_stretched = ((gray_image - min_pixel) /
                      (max_pixel - min_pixel) * 255).astype(np.uint8)

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# Histogram Equalization
equalized_image = cv2.equalizeHist(gray_image)

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# CLAHE
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_image = clahe.apply(gray_image)

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# Display Images
plt.figure(figsize=(5, 4))

plt.subplot(2, 2, 1)
plt.imshow(gray_image, cmap="gray")
plt.title("Original Grayscale")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(equalized_image, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()
plt.show()

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```

```
# Compare Histograms
plt.figure(figsize=(5, 4))

plt.subplot(2, 2, 1)
plt.hist(gray_image.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 2)
plt.hist(contrast_stretched.ravel(), 256, [0, 256])
plt.title("Contrast Stretched Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 3)
plt.hist(equalized_image.ravel(), 256, [0, 256])
plt.title("Equalized Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 4)
plt.hist(clahe_image.ravel(), 256, [0, 256])
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()

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 Roll No.: 31


# Save Enhanced Images
cv2.imwrite("contrast_stretched.jpg", contrast_stretched)
cv2.imwrite("histogram_equalized.jpg", equalized_image)
cv2.imwrite("clahe_image.jpg", clahe_image)

print("Enhanced images saved successfully!")

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# Roll No.: 31

plt.show()

#Observations

#   | Technique              | What to observe                                                                 |
#| ---------------------- | ------------------------------------------------------------------------------- |
#| Original               | Baseline image with its original contrast                                       |
#| Contrast Stretching   | Wider intensity range and improved contrast                                     |
#| Histogram Equalization | Redistribution of intensity values and enhanced overall contrast         |
#| CLAHE                  | Local contrast enhancement, particularly useful when illumination varies |


# QUESTIONS AND ANSWERS:

# """ 1. What is image contrast, and why is it important in image processing?

# Answer:
# Image contrast is the difference in intensity between the dark and bright regions of an image. High-contrast images have clearly distingu
# Contrast is important because it improves the visibility of objects, edges, and details. Better contrast also helps computer vision syste



# 2. Explain the concept of an image histogram. What information does it provide?

# Answer:
# An image histogram is a graphical representation of the distribution of pixel intensity values in an image.
# For a grayscale image, intensity values normally range from 0 to 255, where 0 represents black and 255 represents white.

# A histogram provides information about:

# The distribution of brightness levels.
# Whether an image is dark or bright.
# The contrast of the image.
# The concentration of pixels within particular intensity ranges.

# It is useful for analyzing and improving image quality.



# 3. Differentiate between Histogram Stretching and Histogram Equalization.

# Answer:

# Histogram Stretching  Histogram Equalization
```

```
# Expands the existing intensity range. Redistributes pixel intensities across the available range.
# Improves contrast by mapping minimum and maximum values.  Improves contrast using the cumulative distribution of pixel intensities.
# Usually preserves the general shape of the histogram. Can significantly change the histogram distribution.
# Simple and useful for low-contrast images.  Useful when pixel intensities are concentrated in a limited range.

# In this experiment, histogram stretching was implemented manually, while histogram equalization was performed using OpenCV.

# 4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

# Answer:
# CLAHE stands for Contrast Limited Adaptive Histogram Equalization. It enhances contrast by dividing an image into small regions called ti

# The main difference is:
# Standard Histogram Equalization: Works globally on the entire image.
# CLAHE: Works locally on different regions of the image.

# CLAHE also limits excessive contrast enhancement using a clip limit, which helps prevent noise from being overly amplified.

# Therefore, CLAHE is particularly useful for images having varying illumination or local contrast problems.



# 5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

# Answer:
# Histogram equalization improves the distribution of pixel intensities and increases image contrast. This makes important features such as
# As a result, feature extraction algorithms can identify useful features more effectively, while segmentation algorithms can better distin
# Therefore, histogram equalization can improve the quality of input data before subsequent computer vision operations.



# 6. Mention three real-world applications where histogram equalization is widely used.

# Answer:
# Three real-world applications are:

# Medical imaging – improving the visibility of structures and details in medical images.
# Satellite and remote-sensing images – enhancing features in satellite imagery.
# Low-light photography – improving visibility and contrast in poorly illuminated photographs.

# These applications are consistent with the types of real-world images suggested in the experiment.




# 7. What are the limitations of global histogram equalization?

# Answer:
# The main limitations of global histogram equalization are:

# It considers the entire image rather than local regions.
# It may over-enhance some areas.
# It can amplify noise.
# It may produce unnatural-looking images.
# It may not perform well when different parts of an image have different illumination conditions.

# CLAHE can help address some of these problems by performing enhancement locally and limiting excessive contrast enhancement.




# 8. How does contrast enhancement improve the performance of object detection and recognition systems?

# Answer:
# Contrast enhancement makes important image features such as edges, boundaries, textures, and objects more visible.

# This can help object detection and recognition systems by:

# Improving feature visibility.
# Making object boundaries clearer.
# Improving the distinction between objects and backgrounds.
# Providing better input for feature extraction algorithms.

# Therefore, contrast enhancement can make subsequent computer vision tasks more effective, especially when the original image has poor con




# 9. Compare histogram-based enhancement techniques with brightness adjustment methods.

# Answer:

# Histogram-Based Enhancement Brightness Adjustment
# Changes the distribution of pixel intensities.  Mainly shifts pixel intensities toward brighter or darker values.
# Can improve image contrast. Primarily changes overall brightness.
# Includes histogram stretching, histogram equalization, and CLAHE. Usually involves adding or subtracting an intensity value.
# Can reveal details that are difficult to see in low-contrast images.  May not improve contrast significantly.
# More suitable when intensity distribution needs improvement.  Suitable when the main problem is excessive darkness or brightness.
```

```
# Thus, histogram-based techniques generally provide more control over contrast than simple brightness adjustment.




# 10. Why is CLAHE preferred for medical imaging and low-light image enhancement?

# Answer:
# CLAHE is preferred because it performs local contrast enhancement instead of applying the same enhancement to the entire image.

# This is useful for medical and low-light images because different regions may have different illumination levels. CLAHE can enhance detai

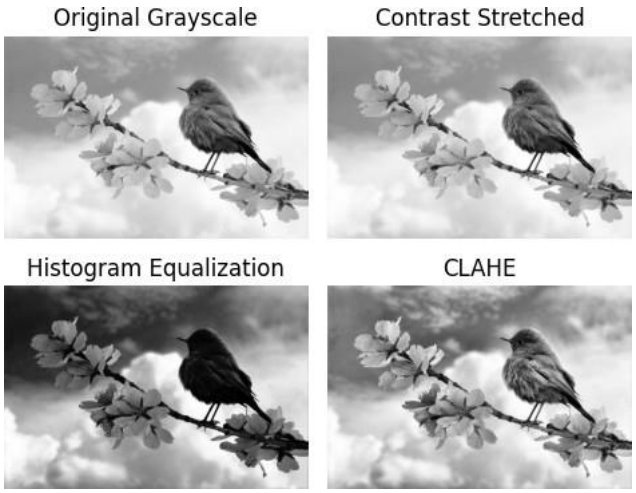
# It also uses a clip limit, which helps reduce excessive amplification of noise.

# Therefore, CLAHE is particularly useful when an image contains varying illumination and important local details need to be enhanced.
# """
```

Choose files bird-8788491_1280.jpg

bird-8788491_1280.jpg(image/jpeg) - 168056 bytes, last modified: 20/08/2026 - 100% done

```
Saving bird-8788491_1280.jpg to bird-8788491_1280 (3).jpg
Image loaded successfully.
Image size: (853, 1280)
```




```
/tmp/ipykernel_929/295114643.py:93: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since M
  plt.hist(gray_image.ravel(), 256, [0, 256])
/tmp/ipykernel_929/295114643.py:99: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since M
  plt.hist(contrast_stretched.ravel(), 256, [0, 256])
/tmp/ipykernel_929/295114643.py:105: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since
  plt.hist(equalized_image.ravel(), 256, [0, 256])
/tmp/ipykernel_929/295114643.py:111: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since
  plt.hist(clahe_image.ravel(), 256, [0, 256])
Enhanced images saved successfully!
```

